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# Effects of overconfidence and dependence on aggregated probability judgments

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## Abstract

**Purpose** – The purpose of this paper is to investigate how expert overconfidence and dependence affect the calibration of aggregated probability judgments obtained by various linear opinion-pooling models.

**Design/methodology/approach** – The authors used a large database containing real-world expert judgments, and adopted the leave-one-out cross-validation technique to test the calibration of aggregated judgments obtained by Cooke's classical model, the equal-weight linear pooling method, and the best-expert approach. Additionally, the significance of the effects using linear models was rigorously tested.

**Findings** – Significant differences were found between methods. Both linear-pooling aggregation approaches significantly outperformed the best-expert technique, indicating the need for inputs from multiple experts. The significant overconfidence effect suggests that linear pooling approaches do not effectively counteract the effect of expert overconfidence. Furthermore, the second-order interaction between aggregation method and expert dependence shows that Cooke's classical model is more sensitive to expert dependence than equal weights, with high dependence generally leading to much poorer aggregated results; by contrast, the equal-weight approach is more robust under different dependence levels.

**Research limitations/implications** – The results suggest that methods involving broadening of subjective confidence intervals or distributions may occasionally be useful for mitigating the overconfidence problem. An equal-weight approach might be more favorable when the level of dependence between experts is high. Although it was found that the number of experts and the number of seed questions also significantly affect the calibration of the aggregated distribution, further research to find the minimum number of questions or experts is required to ensure satisfactory aggregated performance would be desirable. Furthermore, other metrics or probability scoring rules should be used to check the robustness and generalizability of the authors' conclusion.

**Originality/value** – The paper provides empirical evidence of critical factors affecting the calibration of the aggregated intervals or distribution judgments obtained by linear opinion-pooling methods.

**Keywords** Experts, Probability theory, Calibration, Expert aggregation,
Subjective probability distribution, Overconfidence, Dependence

**Paper type** Research paper

## 1. Introduction

Subjective probability assessment has an important role in numerous inferential and decision-making models and is widely used in fields where empirical data is sparse or the cost of data acquisition is prohibitively high. For example, in subjective

This research was partially supported by the ROC National Science Council under grant number NSC 99-2410-H-155-034-MY2. Any opinions, finding, and conclusions or recommendations expressed herein are those of the authors and do not necessarily reflect the views of the sponsors. The authors also thank three anonymous referees for their helpful comments.

Journal of Modelling in Management
Vol. 7 No. 1, 2012
pp. 6-22
© Emerald Group Publishing Limited
1746-5664
DOI 10.1108/17465661211208785

Bayesian inference, elicitation of expert opinion to represent a prior belief about an unknown quantity of interest is a critical step. Likewise, in probabilistic risk analysis, people often rely on subjective probability assessments provided by experts. O'Hagan *et al.* (2006) and Cooke (1991) have reviewed many developments in and uses of expert probability judgments over the years.

We can elicit an expert's state of knowledge and represent the resulting data using a number of different formats. Even for the probability distribution of a variable of interest, the aleatory and epistemic uncertainties of an expert may be summarized in different ways to provide different information and insights. According to Hora (2007) and Önkal and Bolger (2004), the subjective probability interval or distribution estimate is considered to be an informative elicitation format. Several other studies also reveal user preference for interval estimates over point estimates, since prediction intervals provide richer information and may improve the quality of any subsequent decision (Baginski *et al.*, 1993; Chatfield, 2001; Johnson, 1982).

Evaluating probability intervals and the performance of experts at eliciting them has been one of the core themes of previous research regarding interval estimates or forecasts. Numerous calibration studies have assessed the accuracy of probability judgments (Lichtenstein and Fischhoff, 1977; Lichtenstein *et al.*, 1982; Morgan and Henrion, 1990; Russo and Schoemaker, 1992; Cooke, 1991; Shlyakhter *et al.*, 1994). Furthermore, decision makers often have access to more than one expert, and it may be desirable to obtain an aggregated distribution that encapsulates the knowledge of multiple experts. However, few reports focus on the performance of mathematical aggregation methods for probability-distribution or interval judgments. O'Hagan *et al.* (2006) summarize comparison studies of different opinion-pool methods. Cooke and Goossens (2008) review 45 expert-elicitation studies on the performance of specific methods, and report positive results with Cooke's classical model. Several other studies also look at Cooke's data from various perspectives and compare different weighting schemes but conclusive evidence has not yet been found (Wisse *et al.*, 2008; Cooke *et al.*, 2008; Lin and Cheng, 2009). Likewise, Clemen *et al.* (1996) compare two Bayesian aggregation methods and find that not only do both methods outperform intuitive decision makers, but also the more complex Bayesian method does no better than the simpler one. However, other than Kallen and Cooke (2002), Wisse *et al.* (2008) and Cooke *et al.* (2008), few studies have evaluated and compared the effectiveness of different mathematical aggregation methods. Thus, there are few guidelines to help decision makers select appropriate mathematical aggregation methods under different conditions.

The lack of empirical studies investigating the performance of mathematical aggregation methods for probability-distribution judgments is not the only research gap. In the process of eliciting subjective probability judgments, several heuristics are available for use by experts to make judgments about uncertainty, and these heuristics can lead to serious biases. A typical phenomenon, known as expert overconfidence (Lin and Bier, 2008), manifests when distributions or confidence intervals estimated by experts are too narrow. However, the majority of mathematical aggregation methods neither explicitly nor implicitly account for this effect. Instead, they may assume expert estimates to be well calibrated.

Additionally, expert opinions tend to be positively correlated (Meyer and Booker, 2001). If the opinions of positively correlated experts are treated as though they are independent, then the dependence would result in intervals that are overconfident,

and any overlap of their predicted intervals would be taken as stronger evidence of consensus between experts than can be justified (Bier, 2004). Moreover, the value of additional experts decreases when expert opinions are positively correlated. However, few empirical studies in the literature explore how correlation between experts affects the combined intervals.

Thus, in this study, we rigorously check the robustness of some mathematical aggregation methods. We focus mainly on Cooke's classical model and the best-expert approach, and use simple averaging as a benchmark for measuring and comparing performance. We concentrate principally on investigating the effects of expert overconfidence, expert dependence, the number of calibration questions, and the size of expert panels on the aggregated results, to identify the main factors that affect the quality of aggregated judgments. We use a large data set of expert opinions, and conduct cross-validation for out-of-sample evaluation of performance, to verify the effectiveness of various aggregation approaches.

In Section 2 of this article, we discuss the empirical data used in our investigation, proposed scales for measuring expert overconfidence and dependence in the study, the cross-validation approach as a mechanism for unbiased evaluation of performance, and overall performance measures for aggregation models. In Section 3, we first present a preliminary graphical analysis, and then the results of our statistical analysis. In Section 4, we present conclusions and discuss managerial implications.

The Appendix to this paper is a commentary from Roger M. Cooke on the pitfalls of ROAT cross-validation.

## 2. Materials and methods

### *2.1 Linear opinion pools and Cooke's classical model*

Most of the early work on mathematical aggregation of probabilities rested on the technique of opinion pooling, in which an analyst obtain a consensus distribution, $p(\theta)$, as a function of distributions $p_1(\theta), \ldots, p_n(\theta)$ provided by each of a group of $n$ experts (Clemen and Winkler, 2007). This aggregated distribution is then used as an input for subsequent decision making. One popular approach is to use the linear opinion pool:

$$p(\theta) = \sum_{i=1}^{n} w_i p_i(\theta), \tag{1}$$

where $w_i$ is the weight for expert $i$. Although the opinion pool is widely used in real-world applications and is much simpler than the Bayesian or supra-Bayesian approaches (which usually require a decision maker to conduct a sophisticated elicitation exercise), some of the postulated properties required for the combined distribution to be valid might not hold. For example, the probability distributions provided by multiple experts may be positively correlated (Bier, 2004).

The weights $w_i$ assigned in linear opinion pooling usually depend on the decision analyst's beliefs about the experts' expertise. In other words, the weights enable a decision maker to rely more heavily on experts that are perceived to be better. However, a conspicuous shortcoming of an opinion pool is often its inability to model expert performance adequately. Cooke (1991) introduces a widely used and sophisticated method of assigning weights $w_i$ to different experts in opinion pool. In Cooke's model, the weights are determined from experts' performance in a separate exercise that utilizes seed or

calibration variables, i.e. a set of empirical questions in the same subject area for which the values are known (or will be known) by the decision makers, but not by the experts.

A scoring rule is a formula that helps measure the accuracy of a set of judgments or forecasts. The scoring rule used in Cooke's classical model consists of two components, calibration and information. Calibration measures how an expert's probability assessment corresponds to the actual realization, while information measures how the assessed distribution is concentrated. Experts are down-weighted according to how their subjective probabilities depart from the empirical values of the seed variables, or if their assessed distributions are too wide to provide useful information. Let Cal($e$) and Inf($e$) be the calibration score and information score of expert $e$, respectively. In Cooke's model, the weight for expert $e$ is proportional to the product of calibration and information scores:

$$w(e) \propto \text{Cal}(e) \times \text{Inf}(e) \times 1_{\alpha}(\text{Cal}(e)). \tag{2}$$

where $1_{\alpha}(x)$ is an indicator function that establishes a threshold for the minimum calibration score required to receive any weighting in opinion pooling. Thus, a poorly calibrated expert with calibration score $\text{Cal}(e) < \alpha$ would be assigned zero weight. The value of $\alpha$ is usually chosen to maximize the combined score of the resulting aggregated distribution.

Like most scoring rules based on measures of distance or closeness between judged and observed values, both components of the Cooke model are based on the concept of the Kullback–Leibler (K-L) distance, or relative entropy (Kullback and Leibler, 1951). The K-L distance between two discrete probability distributions $P$ and $Q$ is given by:

$$D_{k-L}(P, Q) = \sum_{i} P(i) \log \frac{P(i)}{Q(i)} \tag{3}$$

This measure is always non-negative, and is zero only if $P = Q$.

In Cooke's model, the scoring measures are usually implemented in quantile (percentile) elicitation formats. Experts are requested to state pre-defined quantiles of their subjective probability distributions for a set of random variables. For example, if an expert is requested to provide a median and a 90 percent probability interval for N such quantities, then the 0.05, 0.50, and 0.95 quantiles would divide the range into four inter-quantile intervals and induce a set of probability bins with theoretical probability distribution:

$$P = \{p_1, p_2, p_3, p_4\} = \{0.05, 0.45, 0.45, 0.05\}.$$

The empirical or sample distribution $\boldsymbol{S}(e) = \{s_1, s_2, s_3, s_4\}$ of an expert $e$ can then be found by summing the number of realizations that fall into each probability bin. If expert $e$ is well calibrated, then the sample distribution $\boldsymbol{S}(e)$ will resemble $\boldsymbol{P}$.

The K-L distance or relative entropy of $\boldsymbol{S}(e)$ to $\boldsymbol{P}$ is given by:

$$I(\mathbf{S}(e), \mathbf{P}) = \sum_{i=1}^{4} s_i \log \left( \frac{s_i}{p_i} \right). \tag{4}$$

The expression $2NI(\mathbf{S}(e), \mathbf{P})$ constitutes the so-called likelihood-ratio statistic, which is asymptotically $\chi^2$ distributed with three degrees of freedom. Thus, we define the calibration score for expert $e$ as:

$$\text{Cal}(e) = \mathbb{P}(X \geq 2NI(\mathbf{S}(e), \mathbf{P}) | X\text{-}\chi^2_3) = 1 - \chi^2_3[2NI(\mathbf{S}(e), \mathbf{P})]. \tag{5}$$

Since the familiar $\chi^2$ statistic for goodness of fit is obtained, Cooke's model is also referred to as the "classical model".

In Cooke's model, information represents the degree to which a distribution is concentrated. It is defined by comparing an expert's distribution for each seed variable with respect to a background uniform (or log-uniform) distribution. A more informative distribution would be more concentrated than the uniform distribution. The information score Inf($e$) of an expert $e$ assessing N uncertain quantities is then the average of the K-L distances between the assessed distributions and the background uniform distributions.

*2.2 Data used for evaluating the models*

In this empirical study, we use the same database as Cooke and Goossens (2008). The data set contains 45 expert-judgment studies (using 45 different expert panels) performed by Cooke and colleagues during the period 1990-2006 at the Technical University of Delft and elsewhere. Note that these experts answered important questions within their domains of expertise (instead of almanac or general-knowledge questions). As summarized in Cooke and Goossens (2008), the applications of these expert studies ranged from nuclear power to banking.

For each elicitation question, experts provided at least three percentile points (5, 50 and 95 percent) representing their uncertainty, but not the complete distribution shapes. In each study, the seed questions (or seed variables), for which the true values were known, were chosen to resemble as closely as possible to the questions of interest that the study was to address. The data set included 521 experts and 519 seed questions, for 4,562 subjective probability judgments or probability intervals.

*2.3 Measurement of overconfidence and dependence*

*2.3.1 Measurement of overconfidence*. Several continuous calibration measures might be used to evaluate the performance of interval estimates. For example, the $\chi^2$ calibration score in the Cooke model is based on the concept of classical hypothesis testing. Hora (2004) argues that a hypothesis-based calibration score, based on counting the numbers of times that values fall into intervals with specified probabilities, does not fully use the information contained in an expert's probability judgment. Thus, he proposes that researchers use the discrepancy between the cumulative distribution function (CDF) and the empirical cumulative distribution function (ECDF) to measure lack of calibration. However, the calibration score suggested by Hora might require more quantile estimates for each question, and more realizations for each variable, than we have in our data set.

Using a less mathematically rigorous approach, Soll and Klayman (2004) suggested a different measure for evaluating the magnitude of overconfidence in probability-interval estimates. Soll and Klayman assume that an expert has a corresponding subjective probability density function (SPDF) for each question of interest (i.e. each variable). In our study, we infer experts' SPDF values from their quantile estimates. In other words, we can fit the quantiles to some probability model to estimate the SPDF. Because for most variables (seed questions) in this investigation, only three quantiles were elicited, and because many quantile estimates were skewed, this study required a three-point approximation that functioned for continuous random variables and that could handle both issues. According to Keefer and Bodily (1983), when only median, 0.05 fractile, and 0.95 fractile are available, a bi-triangular approximation of the probability density

function is an appropriate candidate. Although the bi-triangular density function has a jump, or discontinuity, at the median, the approximation seems quite robust. We then calculate the mean expected absolute deviation (MEAD) from various SPDFs and the mean absolute deviation (MAD) by measuring the difference between the realization and the mean (or median) of the SPDF. In this regard, Soll and Klayman offer the following definition:

$$\mathrm{M} = \frac{\mathrm{MEAD}}{\mathrm{MAD}}, \tag{6}$$

where MAD is the average distance between the true value and the mean of the SPDF (and can thus be considered a measure of the ideal interval width), and MEAD is the average deviation calculated based on an expert's SPDF (and thus represents an expert's subjective interval width). This interpretation shows that M is the ratio between the subjective interval size and the ideal interval size. A ratio of $\mathrm{M} < 1$ indicates overconfidence that cannot be attributed to random error or any other source of variability.

This new overconfidence measure was applied to evaluate the overall overconfidence levels for different expert panels, and the results were compared to those obtained from binary calibration measure used in Lin and Bier (2008). The high correlation between these two measures indicates that the new instrument has good criterion-related validity and empirical validity.

*2.3.2 Measurement of dependence*. Rather than measuring the dependence between point estimates, we are dealing with much more complex interval estimates. The choice of appropriate measures of dependence for situations like this is not straightforward. It is possible to use the pairwise correlation measure suggested by Kallen and Cooke (2002), where only an expert's median assessments are considered. In this measure, an expert receives a score of 1 if his or her median assessment is above the true value, and $-1$ otherwise. Because the realizations for the seed questions are known, we can use Pearson's correlation coefficient (product moment correlation) between these 1's and $-1$'s to evaluate the pairwise correlation between experts.

However, the above correlation measure uses only part of the information assessed by the experts. Therefore, we propose using an alternative measure, Kendall's $\tau_b$, a non-parametric measure of association based on the number of concordances and discordances in paired observations. In other words, $\tau_b$ measures the association between the actual bin distributions between two experts. Although this imputation is not perfect, it allows us to measure the correlation of quantiles of the realizations, and to use all the estimates provided by experts. The formula for Kendall's $\tau_b$ is given by:

$$\tau_b = \frac{N_s - N_d}{\sqrt{(N_s + N_d + T_r)(N_s + N_d + T_c)}}, \tag{7}$$

where $N_s$ represents the number of concordant pairs, $N_d$ is the number of disconcordant pairs, and both $T_r$ and $T_c$ are corrections for tied pairs.

*2.4 Performance measures of various aggregation models*

Comparing the probability distributions generated by various mathematical aggregation models with the true values of the events or variables in questions offers a way to compare the accuracy of different aggregation approaches. This is one

of the purposes of scoring rules for awarding scores to experts (or virtual decision makers). To avoid the possibility of favoring Cooke's classical model, we do not use Cooke's scoring rule (i.e. the calibration and information scores based on K-L distance) as a performance measure.

Schervish (1989) suggested that proper scoring rules for individual variables be interpreted in terms of average probabilities. Thus, we use the Brier score (a quadratic score) proposed by Brier (1950) in our investigation. For an assessment of discrete events, let $p_j$ be an expert's (or decision maker's) subjective probability of an event $O_j$, and $d_j = 1$ if event $O_j$ occurs and 0 if $O_j$ does not occur. Then the Brier score is defined by:

$$BS = \sum_{j=1}^{n} (p_j - d_j)^2 \tag{8}$$

The average (arithmetic mean) Brier score, $\overline{BS}$, averaged across various questions within a study is then taken as the performance of an aggregation method. Although $\overline{BS}$ can be further decomposed into different components, we consider only the overall mean score here.

Of course, a universal measure for operational evaluation of probabilistic judgments does not exist. A proper score might still not be able to measure short-run reliability of a series forecasts because a forecaster might still have incentive to distort his opinion (Seidenfeld, 1985). It should also be added that some of the proper scoring rules have been criticized for not being effective.

*2.5 Cross-validation*

Performance scores reported in most of the studies conducted by Cooke are calculated within sample (Cooke, 1991; Cooke and Goossens, 2008). In other words, the method uses the same data to determine both expert weights and performance scores, so results may slightly favor Cooke's method. By using the resampling procedure, we guarantee that the combined distributions for each particular seed question in each study are calculated using weights derived from the other seed variables, and so we provide an unbiased measure of performance. The steps of the (k-fold) cross-validation algorithm are as follows:

(1) For each elicitation study, divide the seed variables into $k$ mutually exclusive subsets.

(2) Set $i = 1$.

(3) Exclude subset $i$.

(4) Use Cooke's method to calculate the performance weights based on the remaining seed variables.

(5) Use the weights obtained in Step 4 to calculate the combined distribution (quantiles) for the seed variables that were excluded in Step 3.

(6) Set $i = i + 1$.

(7) If $i \leq k$, return to Step 3 and repeat.

(8) After collecting the $k$ combined distributions, calculate the score for this set of distributions by using Cooke's scoring rule (including both calibration and information).

In our study, we set $k = N$. Consequently, the algorithm was a leave-one-out procedure.

## 3. Results

### *3.1 Computation results of various measures*

Table I summarizes the data used in our analysis. The table shows measures of overconfidence and dependence for various studies (i.e. expert panels), and records the mean Brier scores for three different models: Cooke's performance weighting (PW), EQ, and the best expert (BE). With regard to the level of overconfidence, the results show that M scores vary a great deal between studies, with values ranging from 0.1 (seriously overconfident) to 1.2 (slightly under-confident). The results also show that M scores for 22 of the 27 studies are lower than 1. In other words, overconfidence exists in most expert panels.

Similarly, Kendall's $\tau_b$, a measure of dependence between experts within the same study, also shows considerable variation. Although average pairwise correlation coefficients for some studies are close to 0, indicating little or no dependence between experts, the average pairwise correlation coefficients for most studies are greater than 0.3. Furthermore, some of the studies show strong positive correlation between expert judgments, with Kendall's $\tau_b$ scores as high as 0.7. All of the associated Kendall's $\tau_b$ scores are positive, supporting our belief that expert opinions tend to be positively correlated. This reflects the fact that most experts in a panel may come from the same

| Expert panel | No. of expert | No. of seed | M | T | Mean probability score BE | EQ | PW |
|---|---|---|---|---|---|---|---|
| 1. Acrylonitrile | 7 | 10 | 0.849 | 0.260 | 0.950 | 0.510 | 0.950 |
| 2. Option trading | 5 | 34 | 1.010 | 0.159 | 0.510 | 0.534 | 0.510 |
| 3. Dike ring | 17 | 47 | 1.045 | 0.802 | 0.587 | 0.595 | 0.587 |
| 4. Crane risk (flanges) | 10 | 8 | 0.521 | 0.307 | 0.810 | 0.510 | 0.610 |
| 5. Crane risk | 8 | 10 | 0.379 | 0.219 | 1.010 | 0.610 | 0.610 |
| 6. Groundwater | 7 | 10 | 0.741 | 0.364 | 0.670 | 0.510 | 0.670 |
| 7. Rocket propulsion | 4 | 13 | 1.105 | 0.272 | 0.941 | 0.572 | 0.695 |
| 8. Space debris | 7 | 26 | 0.299 | 0.706 | 0.818 | 0.602 | 0.818 |
| 9. Composite materials | 6 | 12 | 0.547 | 0.039 | 1.110 | 0.510 | 0.577 |
| 10. Radiation in food | 7 | 6 | 0.735 | 0.655 | 0.643 | 0.643 | 0.643 |
| 11. Dry deposition | 8 | 14 | 0.745 | 0.504 | 0.567 | 0.510 | 0.567 |
| 12. Atmospheric dispersion | 8 | 23 | 0.763 | 0.660 | 0.823 | 0.580 | 0.719 |
| 13. Early health effects | 7 | 15 | 0.482 | 0.544 | 1.123 | 0.617 | 0.830 |
| 14. Radiation dosimetry | 5 | 38 | 0.628 | 0.309 | 0.594 | 0.510 | 0.594 |
| 15. Soil transfer | 4 | 31 | 0.349 | 0.698 | 0.949 | 0.794 | 0.923 |
| 16. Wet deposition | 7 | 19 | 0.712 | 0.308 | 0.678 | 0.510 | 0.552 |
| 17. Gas pipelines | 16 | 14 | 0.133 | 0.304 | 0.681 | 0.624 | 0.624 |
| 18. Montserrat | 10 | 8 | 0.521 | 0.380 | 0.810 | 0.510 | 0.710 |
| 19. Movable barriers | 8 | 14 | 0.313 | 0.396 | 0.939 | 0.624 | 0.796 |
| 20. Real estate | 5 | 31 | 0.780 | 0.550 | 0.600 | 0.510 | 0.600 |
| 21. River dredging | 6 | 8 | 0.914 | 0.011 | 0.710 | 0.510 | 0.510 |
| 22. Sulphur trioxide | 4 | 7 | 1.140 | 0.199 | 0.967 | 0.510 | 0.739 |
| 23. Building temperature | 6 | 48 | 1.045 | 0.369 | 0.660 | 0.510 | 0.560 |
| 24. Atmospheric dispersion (TNO) | 7 | 36 | 0.845 | 0.291 | 0.688 | 0.532 | 0.666 |
| 25. Radioactive deposition (Delft) | 4 | 22 | 1.236 | 0.519 | 0.619 | 0.583 | 0.583 |
| 26. Atmospheric dispersion (Delft) | 11 | 36 | 0.788 | 0.582 | 0.710 | 0.532 | 0.554 |
| 27. Water pollution | 11 | 9 | 0.349 | 0.495 | 0.688 | 0.688 | 0.688 |

**Table I.** Overconfidence, dependence, and performance scores for EQ, PW, and BE over 27 risk-assessment studies

school of thought, or use a similar conceptual model or problem decomposition in making their probability assessments.

Regarding the average Brier score for assessing aggregation-method performance, our calculation results show that for the BE approach, there is wide variation among the average probability scores for different studies. The variations are much smaller for PW and EQ aggregation approaches. Overall, the PW and EQ approaches seem to outperform the BE approach.

Unfortunately, because only 5th, 50th, and 95th percentiles were elicited (and thus a set of probability bins with probability 0.05, 0.45, 0.45, 0.05 is used to denote expert's subjective probability), the Brier score for individual subjective probability estimates has an upper bound of 1.31 and a lower bound of 0.51. This might affect the resolution of this performance measure. Therefore, other scoring rules with greater discriminating power might need to be considered in the future to clarify this issue.

### *3.2 Preliminary graphical analysis*

Figure 1 shows a plot of the magnitude of overconfidence versus the average Brier score for all studies. This figure suggests a negative correlation between level of overconfidence and the average Brier score. Since a high Brier score implies an inferior subjective probability assessment, such a negative correlation suggests that mathematical aggregation models cannot fully counteract the effect of expert overconfidence, especially when the level of overconfidence is high. In other words, even after applying various aggregation approaches to combine expert judgments, studies in which experts are more overconfident can still be expected to achieve lower calibration scores.

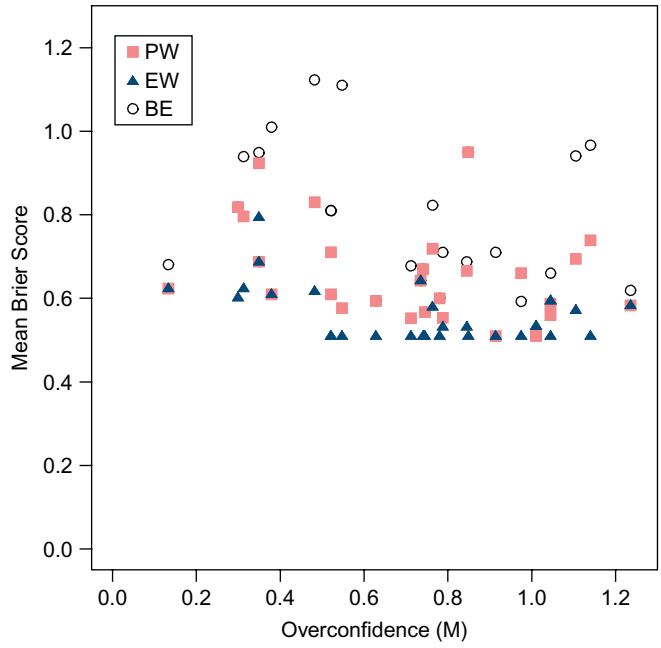



**Figure 1.**
Level of overconfidence versus performance scores for various aggregation models

We also found that aggregations based on EQ and Cooke's PW approaches outperform those obtained via the BE approach and that the EQ approach appears marginally better than the Cooke's classical model. This finding is different from that reported by Cooke and Goossens (2008), but consistent with findings reported by Clemen (2008). Although Cooke (2008) argues that the leave-one-out method is not objective in evaluating expert performance, we would argue that the results presented by Cooke and Goossens (2008) are slightly biased by the fact that Cooke and his colleagues use the same (calibration and information) scoring rules both to determine the weights of the experts and to evaluate the performance of their classical model. Furthermore, by not using an out-of-sample evaluation, bias effects can propagate and become even more significant.

Figure 2 shows the average Brier scores versus average pairwise dependence between experts (as measured using Kendall's $\tau_b$) for each study or expert panel. Different symbols are used to highlight the performance scores of different aggregation methods. The results suggest a positive correlation between level of dependence and the average Brier score. This means that higher dependence between experts will lead to inferior performance, a phenomenon that has been hypothesized, but not previously demonstrated in the literature. However, the correlation between expert dependence and model performance looks weak compared to the correlation between overconfidence and model performance. This may indicate that although expert dependence is an important factor in the quality of expert judgments and may worsen the overconfidence effect as discussed in Bier (2004), opinion-pooling methods perform robustly under various levels of dependence. Thus, while the more sophisticated Bayesian aggregation methods explicitly consider expert dependence, it is not clear that the use of such complex models is justified by the magnitude of the dependence effect.

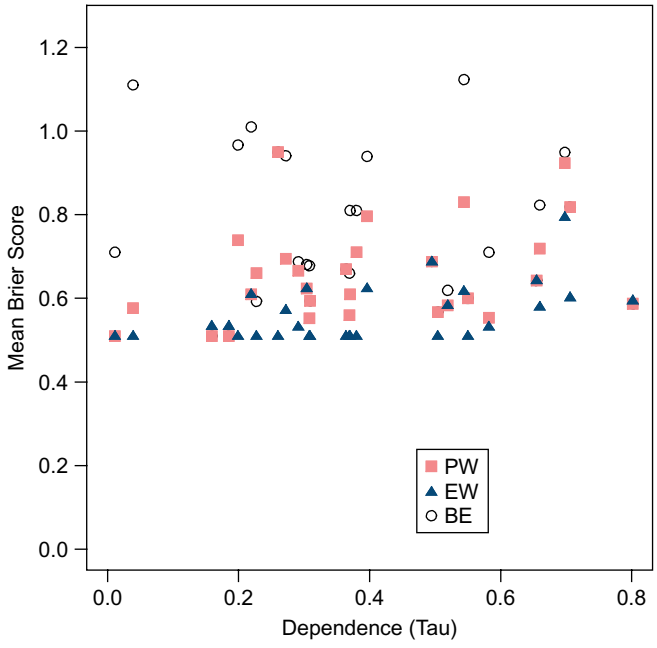



**Figure 2.**
Level of dependence versus performance scores for various aggregation models

*3.3 Model fitting*

Linear models are used to investigate relationships between our independent variables (i.e. different aggregation approaches, characteristics of different studies or expert panels) and the dependent variable (i.e. mean Brier score). Because the focus of our investigation is to determine whether there is a statistically significant difference among the three mathematical aggregation approaches, we obtain an analysis-of-covariance (ANCOVA) interpretation by making the aggregation method a factor. The remaining independent variables, including the overconfidence M, the dependence $\tau_b$, the number of experts, and the number of seed variables, are all treated as covariates. We also treat the interaction terms between the aggregation method and covariates as intervening effects.

Since the number of data points in this analysis is not large, and there is significant noise in both the expert judgments themselves and in the graphical analysis presented above, we thought it prudent to ignore third- or higher order interaction effects. Thus, to find a parsimonious model for representing and interpreting our results, we started with a full second-order model, and then gradually reduced the model by taking out non-significant interaction terms. Our final model contained all five main effects and one interaction term (between expert dependence and the mathematical aggregation method). Because the *Dependence* × *Method* interaction was easily identifiable in our preliminary graphical analysis, we decided to keep this term in the final model, even though it was only marginally significant ($F(2, 78) = 2.13$, $p$-value $= 0.12$). Table II presents the results of our ANCOVA on the mean Brier score for the various mathematical aggregation methods and other covariates. In Table III, we summarize the coefficient estimates, the standard errors of these estimates, and their corresponding $p$-values.

Results in Table II show that the $F$-ratio for the main effect of aggregation methods was statistically significant ($F(2, 78) = 22.19$, $p < 0.01$). Therefore, we conclude that there is a difference between methods for aggregating subjective probability distributions. *Post hoc* multiple comparisons were also conducted to further explore the differences between the mean Brier scores obtained using various methods, after adjusting for the effects of other covariates. We found no significant differences between the PE and EQ approaches. However, both methods outperformed the BE approach, indicating the need for inputs from multiple experts.

With regard to the tests of covariates, the highly significant overconfidence effect ($F(1, 78) = 15.15$, $p < 0.01$) suggests that none of the aggregation approaches investigated in this article effectively counteract the effect of expert overconfidence. In other words, intrinsic expert overconfidence present in the original expert elicitation

| Source of variation | Sums of squares | df | Mean squares | F |
|---|---|---|---|---|
| Method | 0.553 | 2 | 0.276 | $22.19^{***}$ |
| Overconfidence | 0.188 | 1 | 0.188 | $15.15^{***}$ |
| Dependence | 0.009 | 1 | 0.009 | 0.75 |
| Number of experts | 0.084 | 1 | 0.084 | $6.78^{**}$ |
| Number of seeds | 0.066 | 1 | 0.066 | $5.33^{**}$ |
| Method * dependence | 0.057 | 2 | 0.029 | $2.39^{*}$ |
| Error | 0.971 | 78 | 0.012 | |

**Note:** Significant at: $p$-value $^{*} < 0.1$, $^{**} < 0.05$ and $^{***} < 0.01$

**Table II.**
ANCOVA summary table of probability scores

would significantly lower the calibration of an aggregated distribution. Similarly, the existence of expert dependence also affects the quality of the aggregated opinion.

Because there is a marginally significant second-order interaction between aggregation method and dependence, the effects of aggregation method and dependence are non-additive. It is thus more reasonable to look at these two variables simultaneously. The results (Table III) indicate that Cooke's PW model is more sensitive to expert dependence, with higher dependence leading to much poorer aggregation results. On the other hand, the EQ approach is more robust under different dependence levels, even though the calibration score of an aggregated distribution still decreases gradually as the level of dependence increases.

The number of experts and number of seed questions also significantly affect the calibration of the aggregated distribution (Tables II and III). More experts provide a more complete pool of knowledge, and more seed questions increase the possibility of identifying experts with either better substantive (domain) expertise, or better normative expertise (i.e. the ability to adhere to the axioms of probability, and other normative principles such as Bayesian updating and calibration). These findings are also consistent with our expectations.

## 4. Conclusion and discussion

We conducted an out-of-sample evaluation (i.e. leave-one-out cross-validation) to rigorously evaluate the aggregated performance of two major linear opinion-pooling methods (Cooke's classical model and EQ aggregation) and the performance of the best-expert method. We concentrated on exploring the effect of expert overconfidence, expert dependence, the size of the expert panel, and the number of seed questions on the performance of aggregated results, to identify the main factors affecting the calibration of the aggregated interval or distribution judgments. Our statistical analysis found a significant method effect. In particular, both linear opinion-pooling methods outperform the best-expert approach, indicating the need for input from multiple experts.

The significance of the overconfidence effect also suggests that none of the aggregation approaches discussed in this article can effectively counteract the effect of expert overconfidence. Thus, in practice, weighted averages or probabilistic mixtures of multiple experts may not always achieve sufficient calibration. This suggests that methods involving broadening of subjective confidence intervals or distributions may occasionally be useful.

We found a significant second-order interaction between aggregation method and expert dependence in our ANCOVA model. Thus, we need to consider these two

| | Value | SE | *t-value* | *p*-value |
|---|---|---|---|---|
| (Intercept) | 0.842 | 0.056 | 14.902 | 0.000 |
| Method (PW) | − 0.148 | 0.032 | − 4.587 | 0.000 |
| Method (EQ) | − 0.018 | 0.018 | − 1.005 | 0.317 |
| Overconfidence | − 0.154 | 0.047 | − 3.255 | 0.001 |
| Dependence | 0.152 | 0.068 | 2.245 | 0.027 |
| Number of experts | − 0.010 | 0.004 | − 2.558 | 0.012 |
| Number of seeds | − 0.002 | 0.001 | − 2.309 | 0.023 |
| Method (PW) * dependence | 0.129 | 0.073 | 1.761 | 0.082 |
| Method (EQ) * dependence | 0.045 | 0.042 | 1.078 | 0.283 |

**Table III.** Estimate of model coefficients

variables simultaneously. While dependence between experts seems to affect the quality of aggregated distributions for all methods investigated in this study, the results show that Cooke's classical model is more sensitive to expert dependence, with high dependence generally leading to poorer aggregated results. By contrast, the EQ approach is more robust under different levels of dependence. Thus, while these two methods have similar mean Brier scores, the EQ approach might be more favorable in practice, because it shows less variability and is slightly more accurate, especially when the level of dependence between experts is high.

Finally, we found that the number of experts and the number of seed questions also significantly affect the calibration of aggregated distributions. This is consistent with our expectations, because more experts provides a more complete pool of knowledge, and more seed questions increases the discriminatory power of differential weighting approaches. Unfortunately, we have not yet found the minimum number of questions or experts required to ensure satisfactory aggregated performance. Thus, future work might investigate an appropriate threshold for the number of seed questions and experts beyond which robust aggregation results are reasonably assured.

Although the present study has yielded findings that have important managerial implications, the method of investigation is not without problem. The first limitation concerns the leave-one-out cross-validation. Cooke (2008) pointed out possible drawbacks associated with this approach when applied to evaluate probabilistic forecasting schemes. An important area for future research will thus be in refinement of validation method to the analysis. Another problem that arises in data collection has to do with the fact that data in this investigation is mainly on risk assessment. It would be beneficial to replicate this study on larger and different data sets to make definite claims on the effects of overconfidence and dependence on aggregated probability judgments.

Our out-of-sample analysis used only the Brier score as a measure of performance for the various aggregation methods. This is only one of many possible scoring rules available to evaluate these methods. Other metrics include alternative measures of calibration, information, and bias, and/or various decompositions of the various scoring rules. Other measures should therefore be tested to check the robustness and generalizability of our conclusions.

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## Further reading

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## Appendix. Pitfalls of ROAT cross-validation: comment on effects of overconfidence and dependence on aggregated probability judgments

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The authors are applauded for a very clear exposition, and for their innovative use of the TU Delft Expert Judgment data base. It is indeed gratifying to see the discussion move beyond metaphysics and grapple with data. The analysis in this paper builds on Clemen (2008). Using Remove-One-at a –Time (ROAT) sampling, Clemen (2008) found that the classical or performance weight (PW) model outperformed equal weighting (EQ) on only nine of 14 cases, which was not significant. The present paper extends this analysis. This comment shows that ROAT, though suitable in some contexts, is massively wrong for probabilistic combination schemes with likelihood driven weights.

The central question is out-of-sample validation. True out-of-sample validation would involve initializing the PW model on a set of seed variables, and using this model to predict variables of interest, which are later observed. We use expert judgment because the variables of interest are NOT observable, nonetheless there is a handful of studies in which true out-of-sample validation is possible, as reported in (Cooke and Goossens, 2008). Absent observation of variables of interest, the next best option for out of sample validation consists in splitting the seed variables into two equal halves, initializing the model on one half, and using this to predict the other half, and conversely. There are 13 studies in the TU Database with at least 16 seed variables, for which this option was available (8 calibration variables is the minimum we use for the classical model). This is a true out-of-sample validation, although it does not validate performance on unobserved variables of interest. As reported in Cooke (2008), in 20 of the 26 cases the PW outperformed EQ. The probability of seeing 20 or more "successes" on 26 trials, if the probability of success were 0.5, is 0.001247. This refutes the hypothesis that PW is statistically equivalent with EQ, and raises questions about the bias of ROAT.

ROAT sampling is NOT out-of sample validation. Each seed variable is removed one at a time, the model is re-initialized on the remaining seed variables and used to predict the removed variable. Each prediction is made by a "different" model. To appreciate how much these models may differ, Table AI gives the 23 different performance weights for the eight experts that arise as the 23 seed variables in the Eudisp case are removed one at a time. Evidently, the differences between ROAT and true out-of-sample validation can be substantial. The weight for expert 4 varies from 1 to 0.4; that of expert 5 from 0 to 0.59. This is a consequence of the well know volatility of the calibration score, which is observed on every robustness analysis of every study. As pointed out in Cooke (2008), the recalculation of weights when item j is removed tends to give more weight to experts who assessed item j badly, and hence this volatility works "against" the PW model. That of course explains the differences between the results reported in Clemen (2008) and Cooke (2008).

To better understand this bias, suppose two experts state the probability of heads. Let $P_1 = 0.8$ and $P_2 = 0.2$ be the probability of heads for experts 1 and 2. Suppose that the decision maker's probability for heads is a weighted combination of the experts' probabilities, $P_{dm} = wP_1 + (1 - w)P_2$, where the weight of each expert, given observed data, is proportional to the likelihood of each expert's distribution, given the data[1]. After observing n "1"'s and m "0"'s, the experts' likelihood ratio is:

$$\frac{0.8^n \times 0.2^m}{0.2^n \times 0.8^m} = 0.8^{n-m} \times 0.2^{m-n}.$$

If $m = n$, then the weight ratio is 1, and $w = 1/2$. If $m = n - 1$ then the weight ratio is 4 and $w = 4/5$. Thus, if we remove one "0", re-initialize our model and predict the "0" which was removed, we find

| Removed variable | Expert 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|---|
| None | 0 | 0 | 0 | 0.7683 | 0.2317 | 0 | 0 | 0 |
| 1 | 0 | 0 | 0 | 0.8086 | 0.1914 | 0 | 0 | 0 |
| 2 | 0 | 0 | 0 | 0.7928 | 0.2072 | 0 | 0 | 0 |
| 3 | 0 | 0 | 0 | 0.8071 | 0.1929 | 0 | 0 | 0 |
| 4 | 0 | 0 | 0 | 0.7303 | 0 | 0.2697 | | |
| 5 | 0 | 0 | 0 | 0.4094 | 0.5906 | 0 | 0 | 0 |
| 6 | 0 | 0 | 0 | 0.7022 | 0.2978 | 0 | 0 | 0 |
| 7 | 0 | 0 | 0 | 0.6777 | 0.3223 | 0 | 0 | 0 |
| 8 | 0 | 0 | 0 | 0.7928 | 0.2072 | 0 | 0 | 0 |
| 9 | 0 | 0 | 0 | 0.806 | 0.194 | 0 | 0 | 0 |
| 10 | 0 | 0 | 0 | 0.8645 | 0.1355 | 0 | 0 | 0 |
| 11 | 0 | 0 | 0 | 0.7003 | 0.1638 | 0 | 0 | 0.1359 |
| 12 | 0 | 0 | 0 | 0.7042 | 0.1632 | 0 | 0 | 0.1325 |
| 13 | 0 | 0 | 0 | 0.7659 | 0.2341 | 0 | 0 | 0 |
| 14 | 0 | 0 | 0 | 0.6996 | 0.1654 | 0 | 0 | 0.135 |
| 15 | 0 | 0 | 0 | 0.6287 | 0.1637 | 0 | 0.07593 | 0.1317 |
| 16 | 0 | 0 | 0 | 0.704 | 0.296 | 0 | 0 | 0 |
| 17 | 0 | 0 | 0 | 0.6996 | 0.1655 | 0 | 0 | 0.1349 |
| 18 | 0 | 0 | 0 | 0.6286 | 0.1638 | 0 | 0.07588 | 0.1317 |
| 19 | 0 | 0 | 0 | 0.704 | 0.296 | 0 | 0 | 0 |
| 20 | 0 | 0 | 0 | 0.6499 | 0.1537 | 0 | 0.07101 | 0.1254 |
| 21 | 0 | 0 | 0 | 0.5016 | 0.1307 | 0 | 0 | 0.3677 |
| 22 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 |
| 23 | 0 | 0 | 0 | 0.4094 | 0.5906 | 0 | 0 | 0 |

**Table AI.**
Weights for ROAT in Eudisp

that the $P_{dm}$ used for prediction is $P_{dm} = (4/5) \times 0.8 + (1/5) \times 0.2 = 0.68$. Removing one "0", strongly tilts the model toward expert 1, and our prediction probability for heads is 0.68. At the same time we evaluate this model on the "0" which we removed, hence the likelihood for this model on this observation is 0.32. The same holds, mutatis mutandis, when we remove a "1". Suppose we observe $n = m$; then the PW model would use $w = 1/2$. If we truly validated out of sample with $n = m$ fresh observations, the PW likelihood would be $0.5^{2n}$, but the ROAT value would be $0.32^{2n}$. ROAT punishes PW relative to EQ by a factor $(0.32/0.5)^{2n}$. The classical model is more complex than this simple probabilistic model, but the same behavior can be observed in simple artificial examples[2].

**Notes**

1. Such likelihood weights are not proper scoring rules, and do not account for informativeness, nonetheless there is a strong analogy with the classical model, as the driving term in that model is the likelihood of the hypothesis that an expert is well-calibrated.
2. Take expert 1 (2) with 5, 50 and 95 percentiles equal to 0, 4, 8, (2, 6, 10). Take 3 realizations $= 0.5$, 3 realizations $= 9.5$, 2 realizations $= 4$ and 2 realizations $= 6$. PW and EQ coincide and hence should be identical on out of sample validation; however, the score of PW under ROAT is a factor 105 lower than that of EQ.

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